Microsoft

AI for Good | Tech Workshop

Building Agents w/ Microsoft Copilot Studio and Foundry

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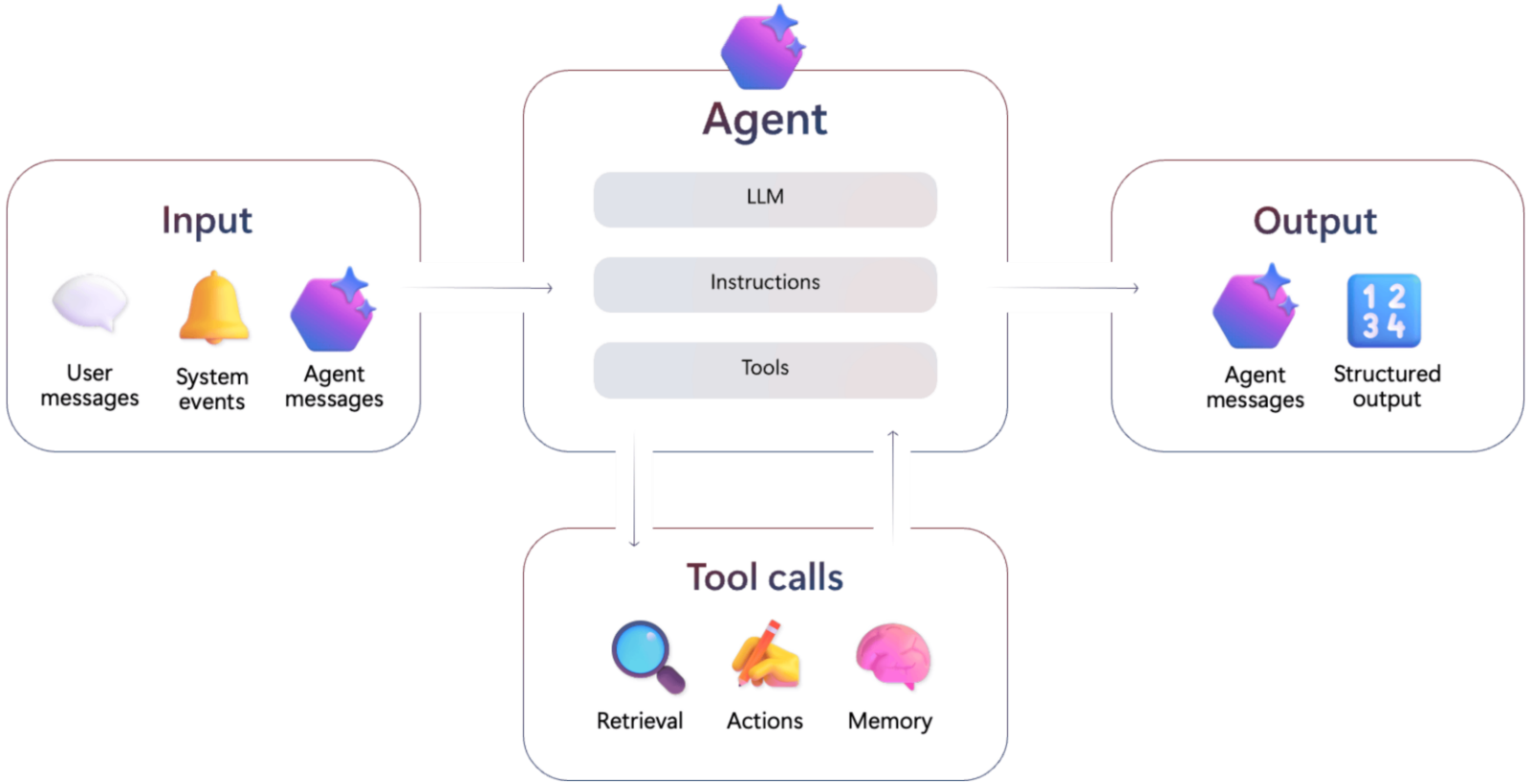
Agenda

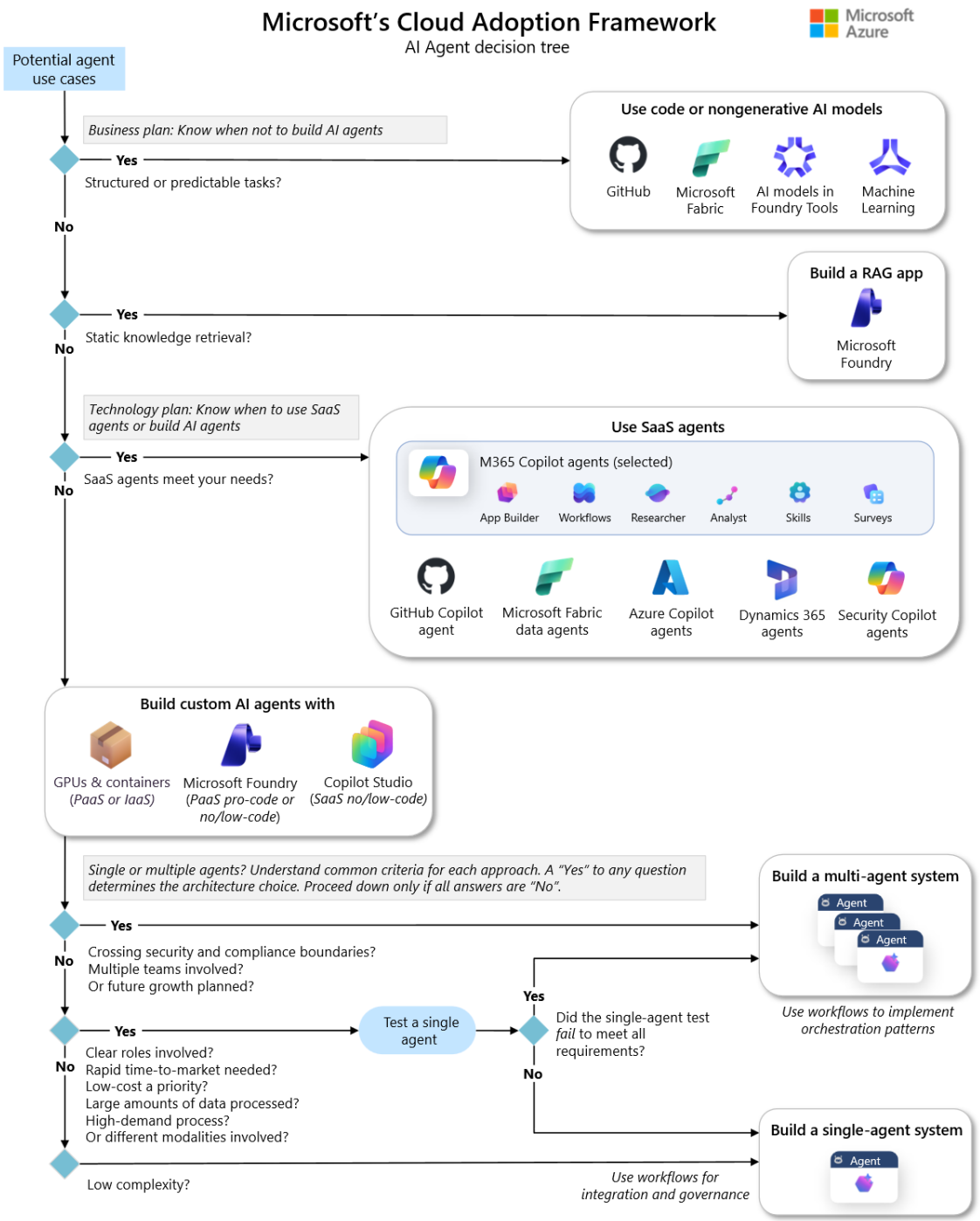
- What is an AI agent?
- When to (and not to) use an AI Agent?
- Agent orchestration patterns
- How can we build agents?
  - Microsoft Copilot Studio
  - Microsoft Foundry
- Use Case: Foodlink
- Workshop
- Q&A

What is an AI Agent?



Reference: [AI Agent Adoption Guidance for Organizations]

When to (and not to) use an AI Agent?



Reference: [Business Plan for AI Agents]

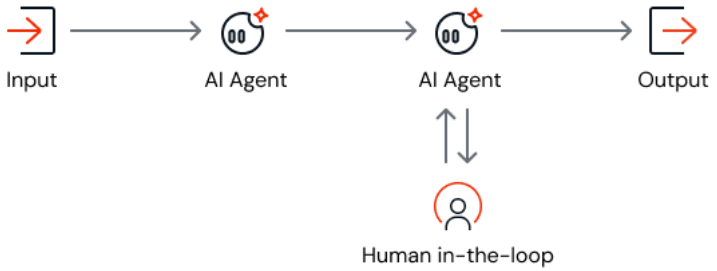
Agent orchestration patterns

Sequential agent pattern

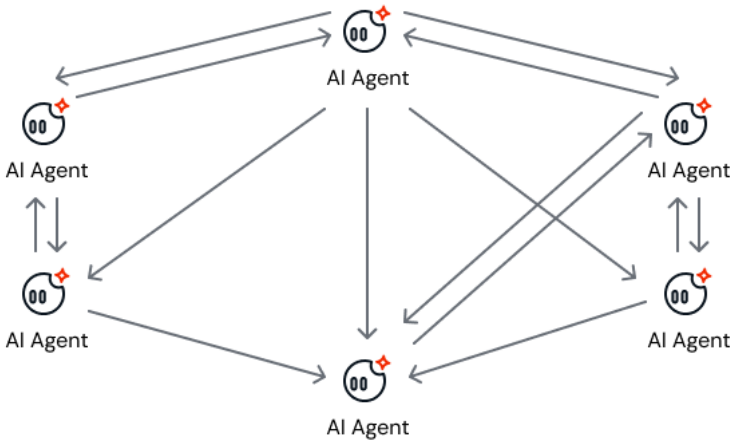
Loop agent pattern

Aggregator agent pattern

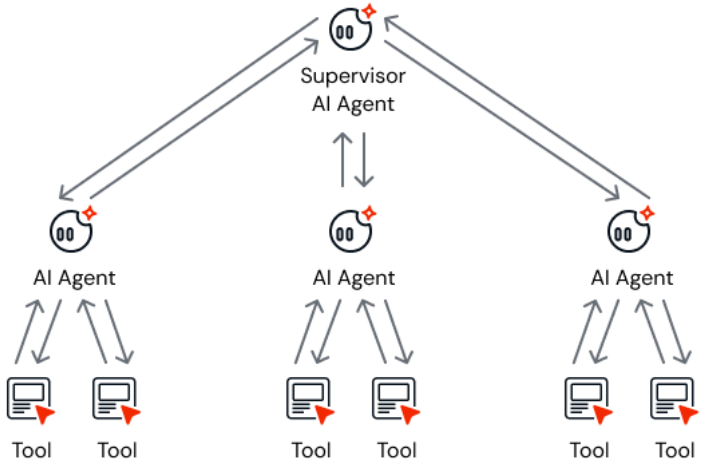
Human-in-the-loop pattern



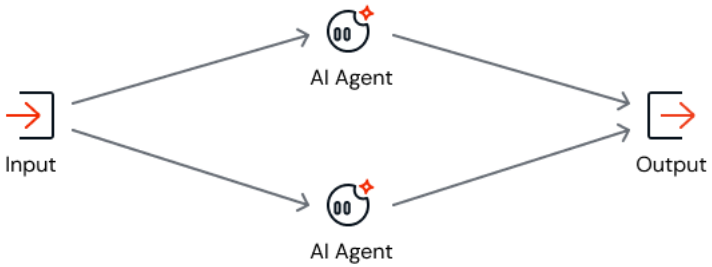
Network agent pattern



Supervisor agent pattern



Parallel agent pattern



Reference: [Agentic patterns - ODC Documentation]

How can we build agents?

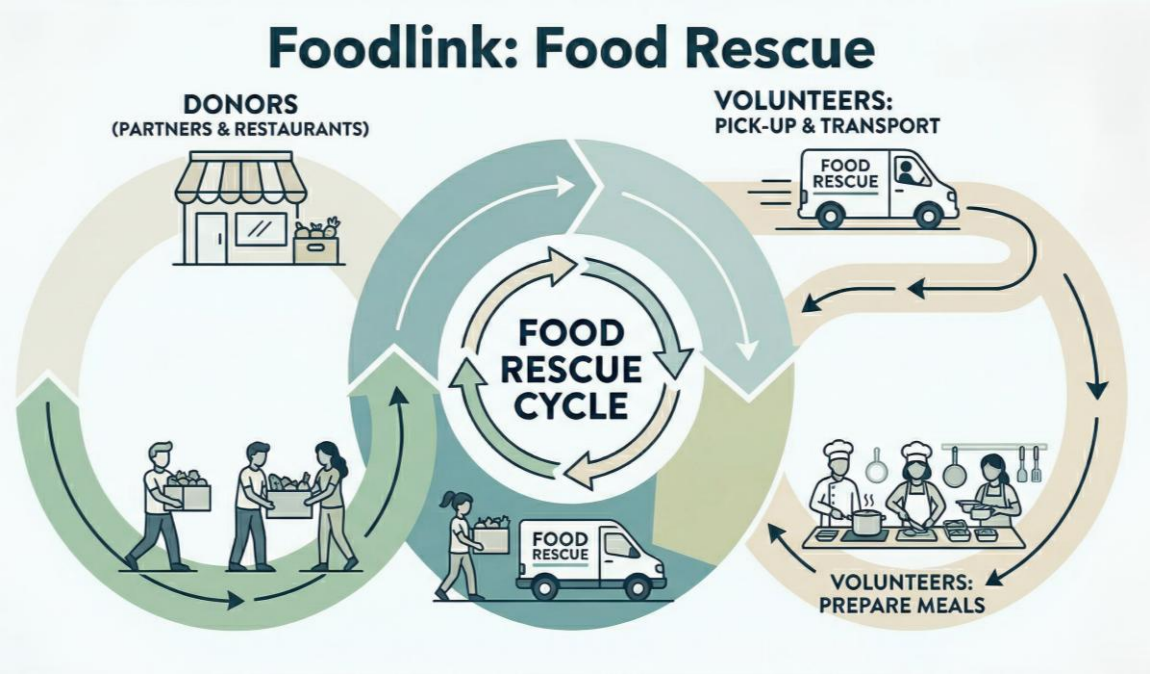
Ease-of-use and development speed

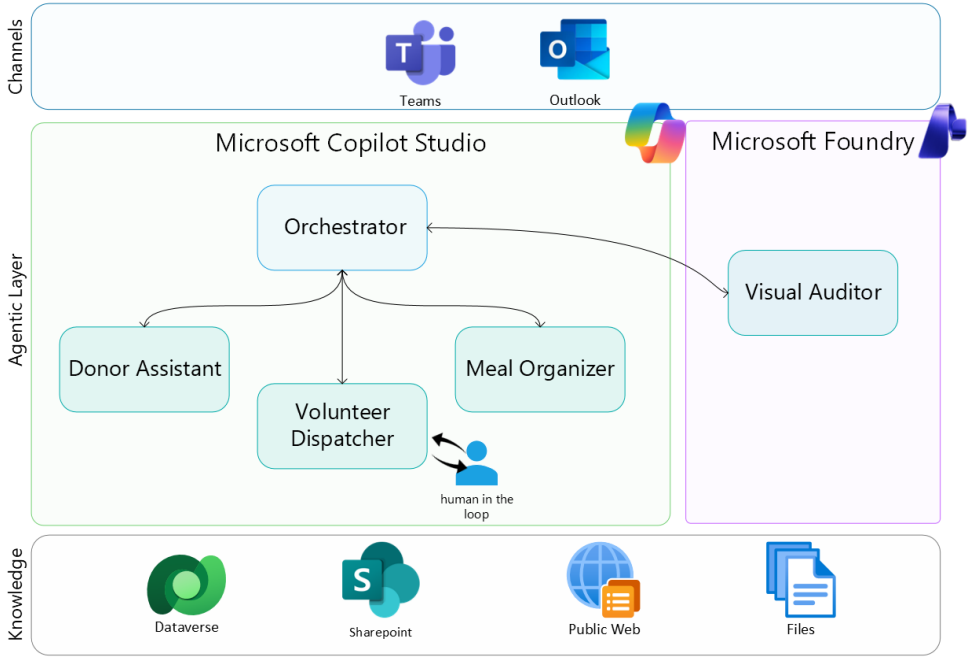
GPUs & Containers Serverless or self-managed GPUs	Microsoft Foundry Agent Service	Copilot Studio Agents
Tool calling orchestration		
Full control	**Managed orchestration or Choose an agent framework Deterministic workflows**	**Fully-managed**
Conversation thread storage		
Custom storage	**BYO-storage** Cosmos DB	**Fully-managed**
Pricing		
Compute	**Per token**	**Per message**
Development environment		
VS Code + GitHub	**VS Code +GitHub + Foundry portal**	**Copilot Studio**

Control and customization

Reference: [Technology Plan for AI Agents]

Use Case: Foodlink





FoodLink serves as an example of a food redistribution network:

- **Donors** (including restaurants and farms) supply surplus food.
- **Volunteers** handle the transportation of goods to local centers.
- **Hubs** transform the ingredients into meals for distribution.

The architecture includes five agents, each with its own tasks and instructions, operating in an **supervisor** and **human-in-the-loop** orchestration pattern**:**

- 1 orchestrator: **FoodLink Agentic AI**.
- 3 child agents: **Donor Assistant**, **Volunteer Dispatcher**, **Meal Organizer**.
- 1 connected Microsoft Foundry agent: **Visual Audit**.

Workshop

👇 **Full guide** 👇

https://github.com/manuelpsnunes/ai-for-good-agent-pathway

Q&A

Thank you

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